IGCSE 0607 Practice Questions — E7.2

Right-angled triangles

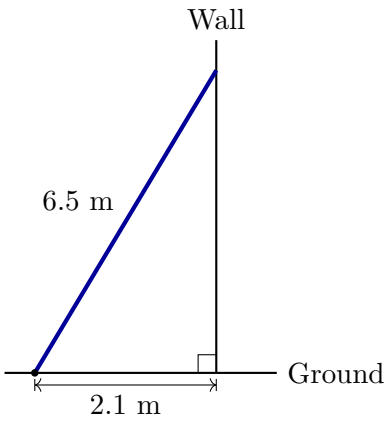
Total Questions: 9 — **Total Marks:** 62 Marks

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  → **Paper 4 (Calculator)** *(9 Qs)*

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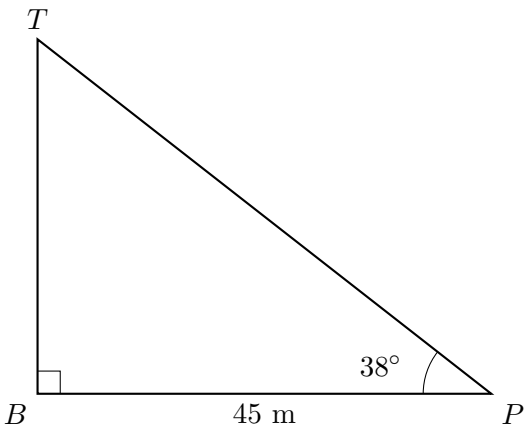
1 A straight ladder of length 6.5 m leans against a vertical wall.
The base of the ladder is 2.1 m from the bottom of the wall on horizontal ground.



Calculate the angle the ladder makes with the horizontal ground.

..... ° **[3]**

2 A surveyor stands at a point P, which is 45 m away from the base, B, of a vertical tower TB. The angle of elevation of the top of the tower, T, from P is 38°.



(a) Calculate the height of the tower, TB.

..... m **[2]**

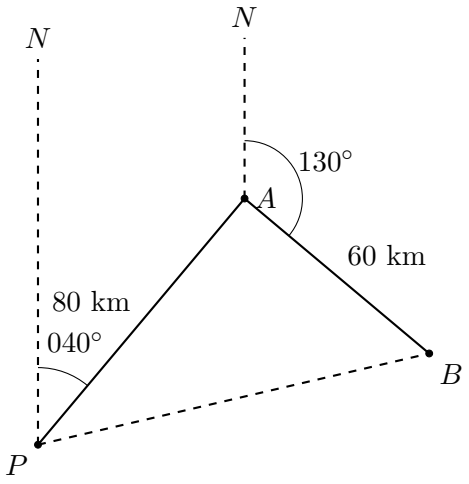
(b) Calculate the straight-line distance PT.

..... m **[2]**

(c) A bird sits at a point X on the tower, where $BX = 20$ m.
Calculate the angle of elevation of the bird from P.

..... ° **[3]**

3 A ship sails from port P on a bearing of 040° for 80 km to point A.
It then changes course and sails on a bearing of 130° for 60 km to point B.



(a) Calculate the direct distance from P to B.

$PB =$ ..... km **[3]**

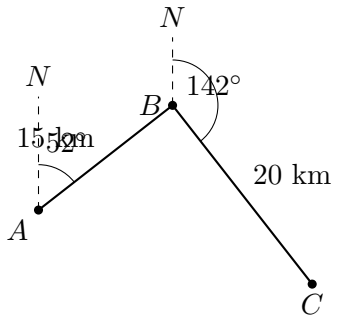
(b) Calculate the bearing of B from P.

Bearing = ... ° **[3]**

(c) Calculate the shortest distance from point A to the path PB.

..... km **[3]**

4 A ship sails from port A to point B on a bearing of 052° for 15 km.
It then sails from point B to point C on a bearing of 142° for 20 km.



(a) Show that angle $ABC = 90°$.

[2]

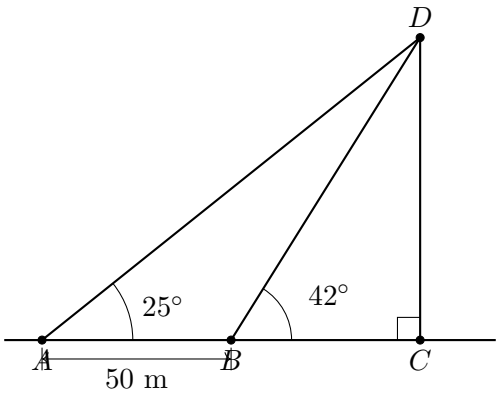
(b) Calculate the distance AC.

..... km **[3]**

(c) Calculate the bearing of C from A.

..... ° **[4]**

5 Point A and point B are on horizontal ground, in a straight line with the base C of a vertical building CD.
The distance between A and B is 50 m.
The angle of elevation of the top of the building, D, from A is 25° and from B is 42°.



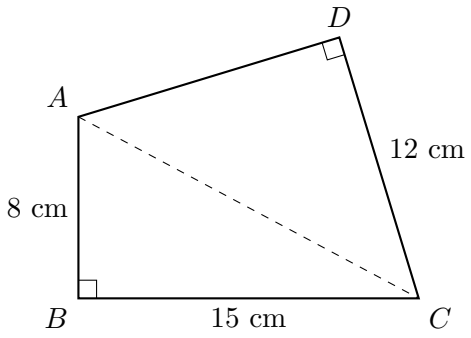
(a) Calculate the height of the building CD.

$CD =$ ..... m **[5]**

(b) A bird flies directly along the straight path from A to D.
Calculate the shortest distance from point B to the flight path AD.

..... m **[3]**

6 The diagram shows a quadrilateral $ABCD$.
Angle $ABC = 90°$ and angle $ADC = 90°$.
$AB = 8$ cm, $BC = 15$ cm, and $CD = 12$ cm.



(a) Calculate the length of AC.

..... cm **[2]**

(b) Calculate the length of AD.

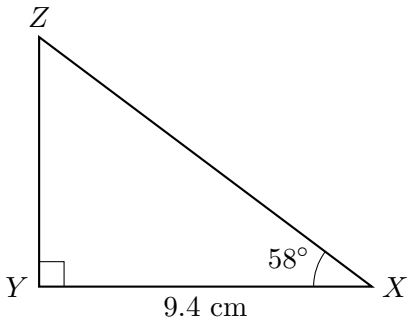
..... cm **[3]**

(c) Calculate the shortest distance from D to the line AC.

..... cm **[3]**

(d) Calculate the size of angle DAB.

..... ° **[4]**



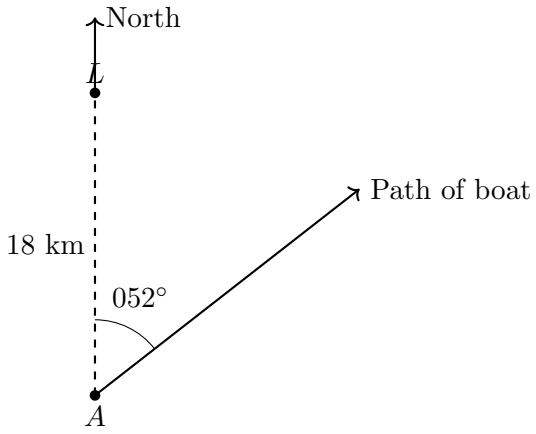
NOT TO SCALE

7 Triangle XYZ is right-angled at Y.
Angle $ZXY = 58°$ and $XY = 9.4$ cm.

Calculate the length of XZ.

..... cm **[3]**

8 A boat sails from a point A in a straight line on a bearing of 052°.
A lighthouse, L, is located 18 km due North of A.



NOT TO SCALE

(a) Calculate the shortest distance from the lighthouse L to the path of the boat.

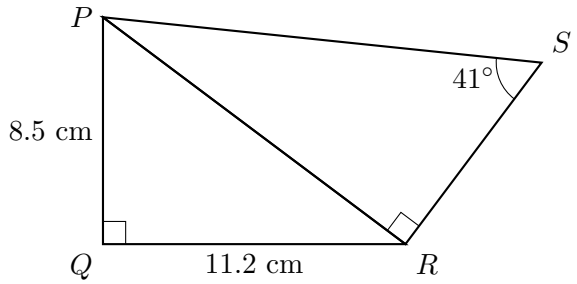
..... km **[2]**

(b) The boat travels at a constant speed of 14 km/h.

Calculate the time it takes for the boat to reach the point on its path closest to the lighthouse.
Give your answer in minutes, correct to the nearest minute.

..... minutes **[3]**

9 The diagram shows a quadrilateral $PQRS$ formed by two right-angled triangles, PQR and PRS.
Angle $PQR = 90°$ and angle $PRS = 90°$.
$PQ = 8.5$ cm, $QR = 11.2$ cm and angle $PSR = 41°$.



NOT TO SCALE

(a) Calculate the length of PR.

..... cm **[2]**

(b) Calculate the length of PS.

..... cm **[2]**

(c) Calculate the perpendicular distance from R to the line PS.

..... cm **[2]**